

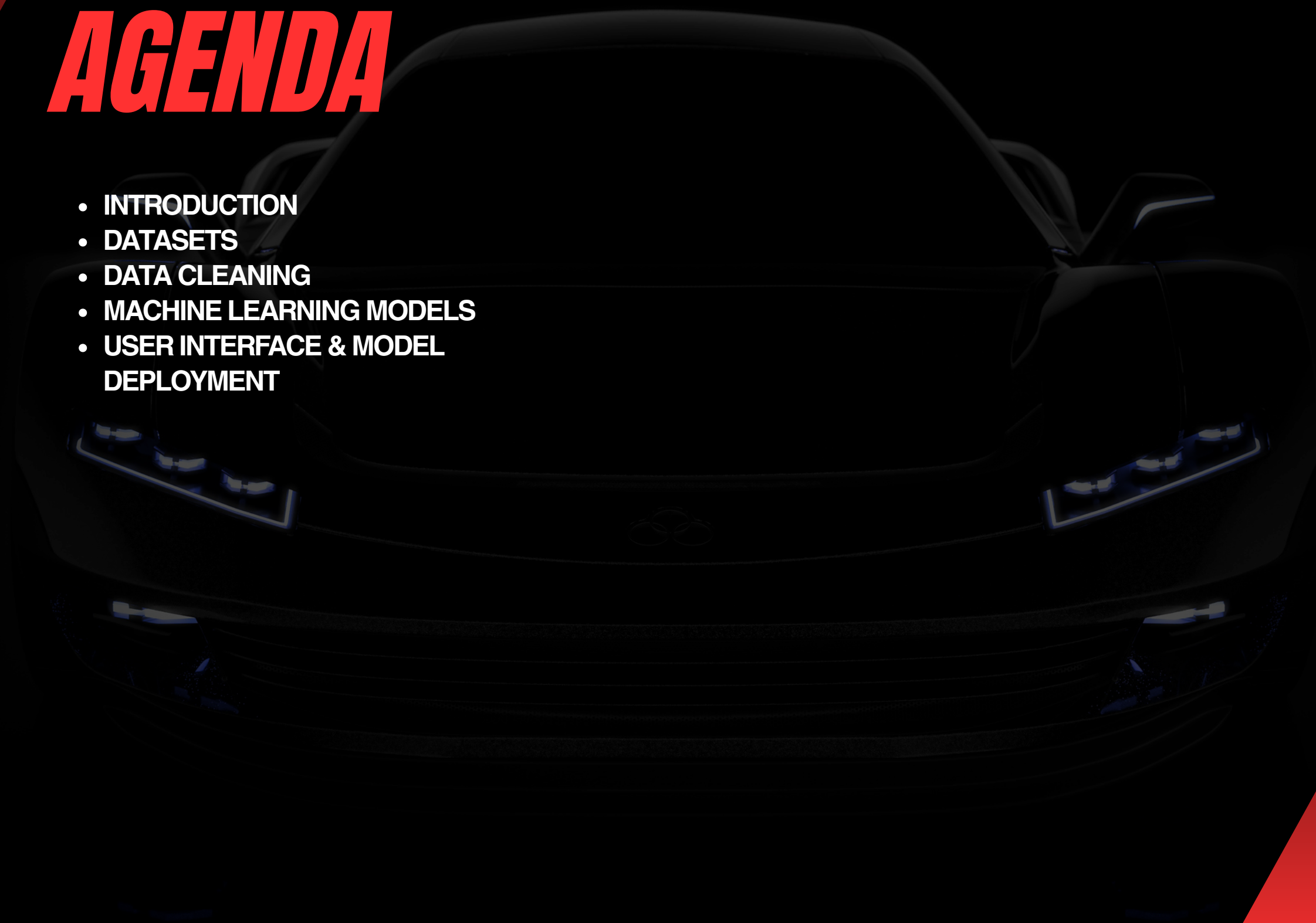
 SALFORD & CO.

AUTOCAR

BEST DEALER AND AUTOMOTIVE

AGENDA

- INTRODUCTION
- DATASETS
- DATA CLEANING
- MACHINE LEARNING MODELS
- USER INTERFACE & MODEL
DEPLOYMENT



INTRODUCTION

In 2024, Uber handled nearly 150,000 bookings. But here's the challenge: 1 in 4 rides was cancelled. Each cancelled trip isn't just a number—it's wasted time, lost trust, and missed opportunities. Let's explore the story behind these cancellations.

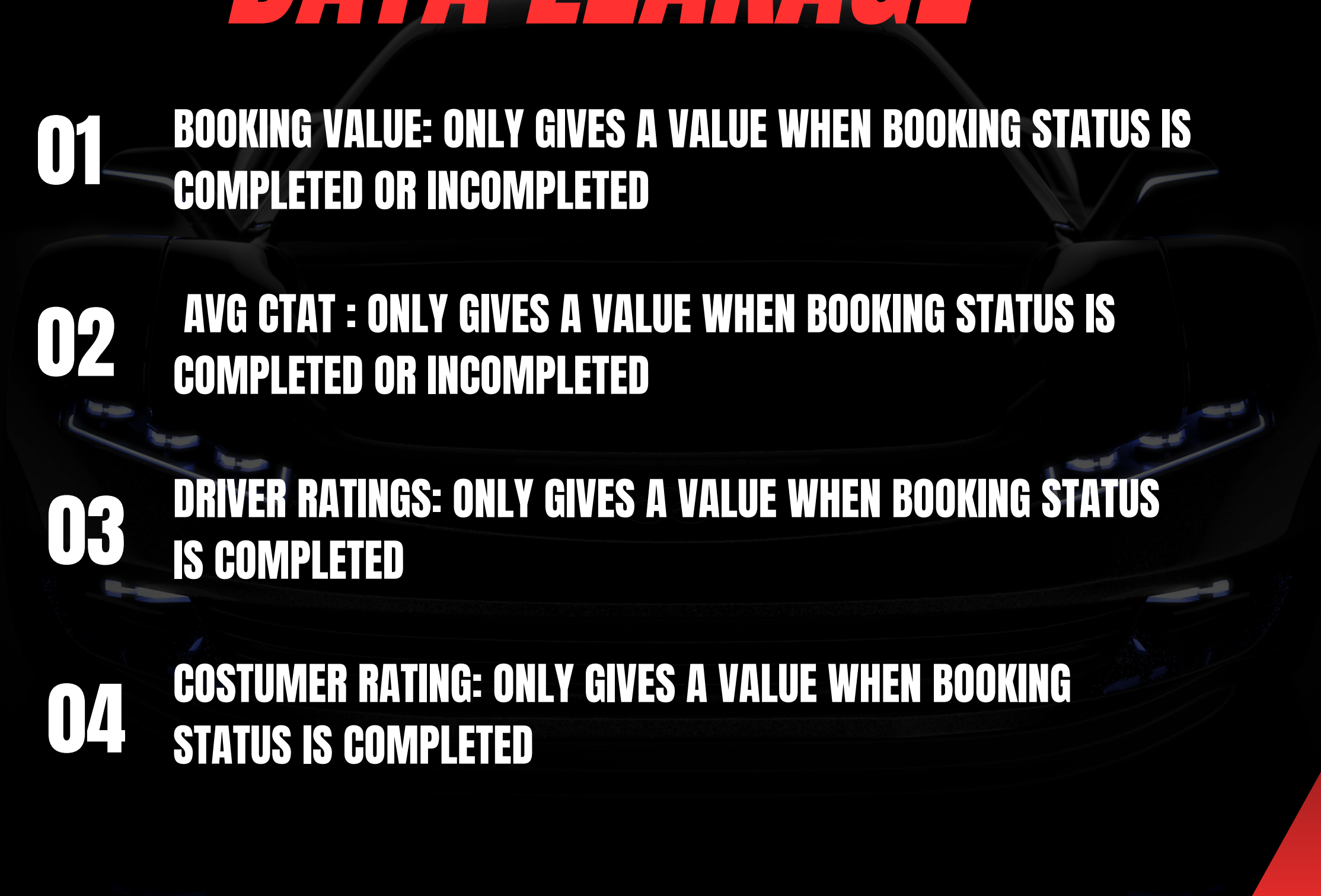


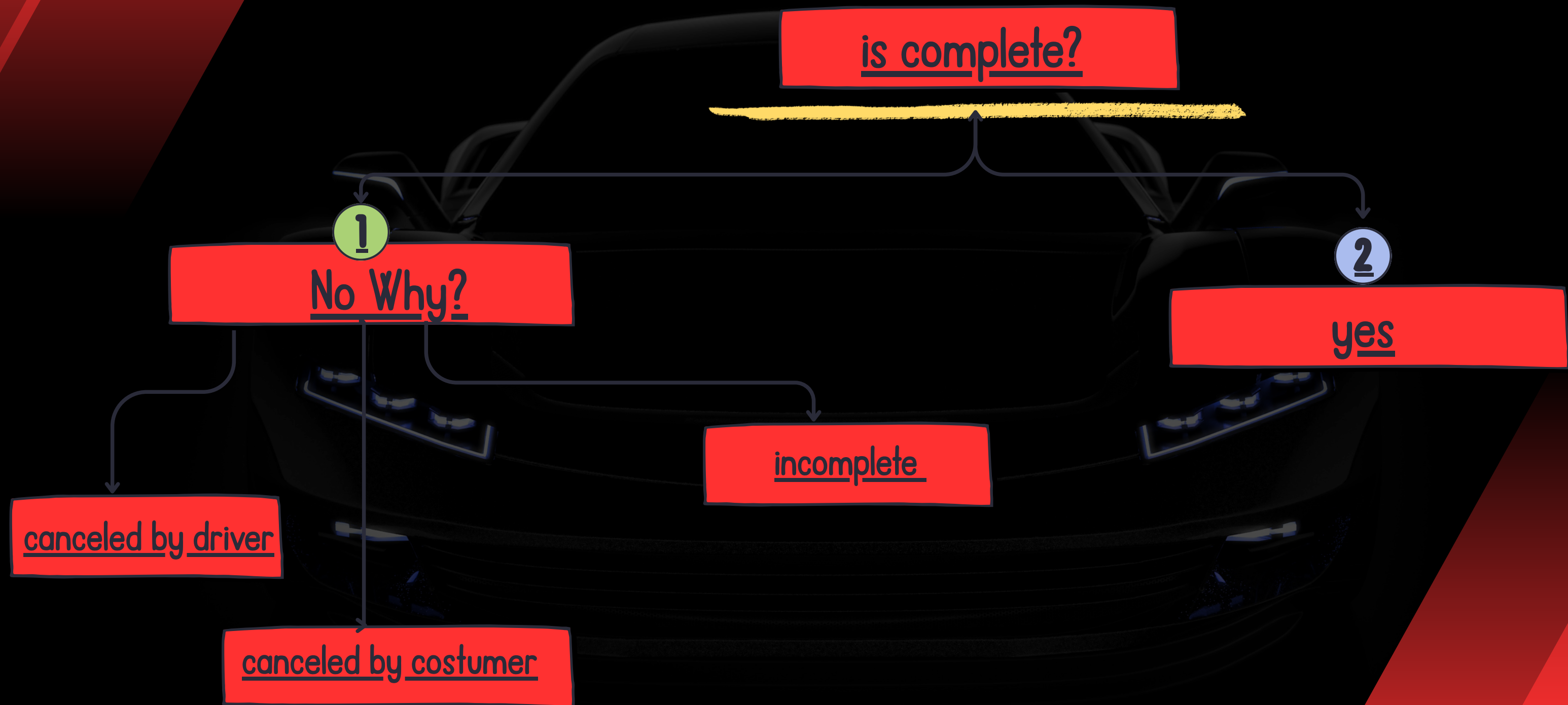
DATASETS

**THE DATASET CAPTURES 150K TOTAL BOOKINGS
ACROSS MULTIPLE VEHICLE TYPES AND PROVIDES
A COMPLETE VIEW OF RIDE-SHARING OPERATIONS
INCLUDING SUCCESSFUL RIDES, CANCELLATIONS,
CUSTOMER BEHAVIORS, AND FINANCIAL METRICS.**



DATA LEAKAGE

- 
- 01** **BOOKING VALUE: ONLY GIVES A VALUE WHEN BOOKING STATUS IS COMPLETED OR INCOMPLETED**
 - 02** **AVG CTAT : ONLY GIVES A VALUE WHEN BOOKING STATUS IS COMPLETED OR INCOMPLETED**
 - 03** **DRIVER RATINGS: ONLY GIVES A VALUE WHEN BOOKING STATUS IS COMPLETED**
 - 04** **COSTUMER RATING: ONLY GIVES A VALUE WHEN BOOKING STATUS IS COMPLETED**



DATA CLEANING

01 HANDLING MISSING VALUES

02 DUPLICATE

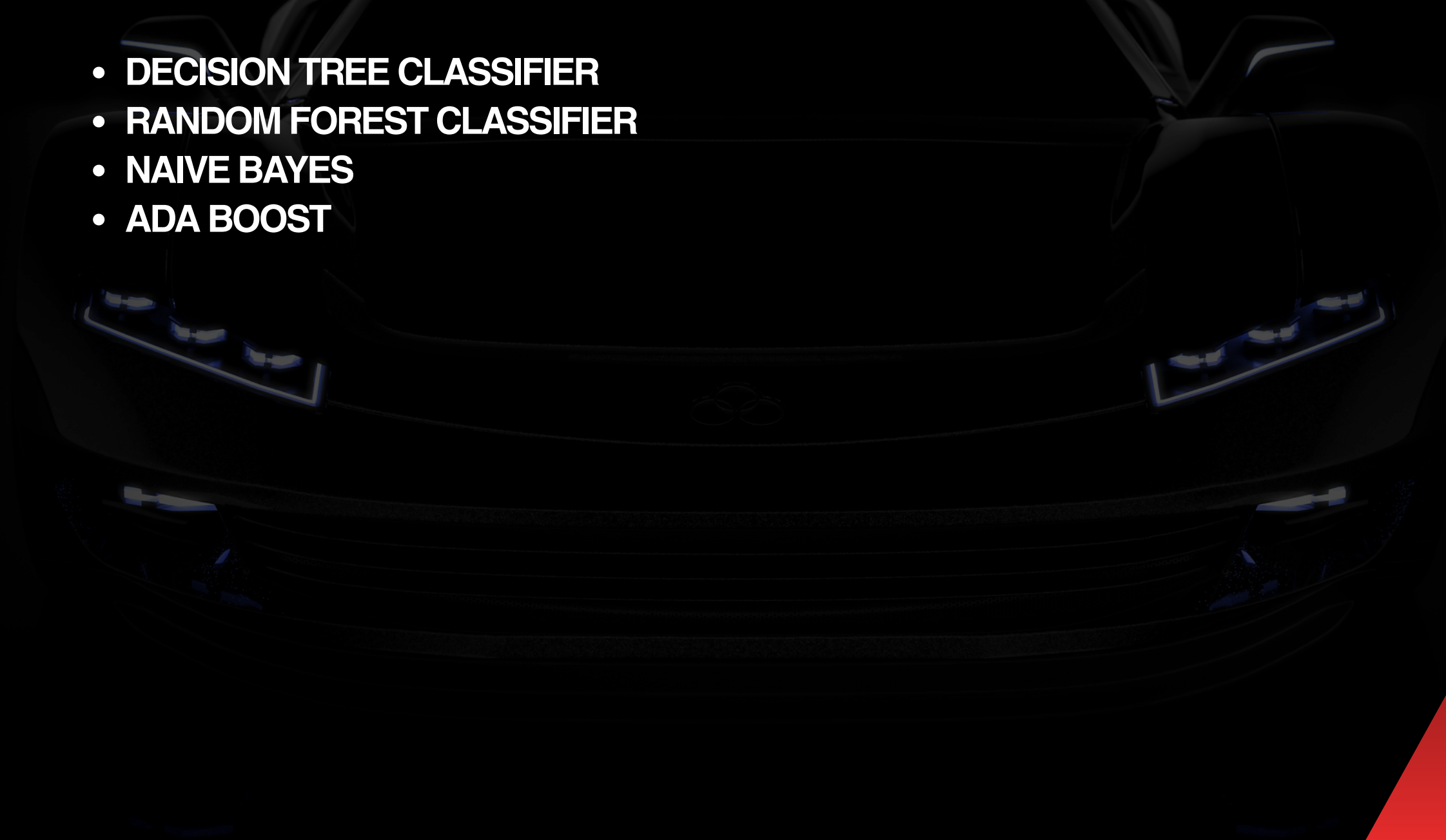
03 DATA TYPES

04 DROP IRRELEVANT COLUMNS



MACHINE LEARNING MODELS

- **DECISION TREE CLASSIFIER**
- **RANDOM FOREST CLASSIFIER**
- **NAIVE BAYES**
- **ADA BOOST**



USER INTERFACE & MODEL DEPLOYMENT

Interactive UI with Gradio: Integrated a Gradio-powered web interface for real-time model inference, allowing users to input ride parameters and instantly view prediction results and cancellation drivers.





THANK YOU